The Hodge theory of Lagrangian fibrations, with applications to hyper-Kähler geometry

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1. Introduction and background

Compact Kähler manifolds with trivial canonical bundle, the higher-dimensional analogues of elliptic curves and K3 surfaces, are central objects in algebraic geometry and mathematical physics. By the Beauville–Bogomolov decomposition theorem [Bea83, Bog74], every such manifold is, up to a finite étale cover, a product of complex tori, Calabi–Yau manifolds, and *hyper-Kähler manifolds*: simply connected compact Kähler manifolds X whose holomorphic 2-forms are spanned by a single symplectic form σ. The decomposition extends to varieties with mild (klt) singularities [BGL22], with hyper-Kähler manifolds replaced by their singular analogues, irreducible symplectic varieties.

Hyper-Kähler manifolds are governed by Hodge theory to a remarkable degree: by the global Torelli theorem [Ver13], a deformation class is described by an explicit arithmetic quotient of a period domain, so a single example determines the whole class. The fundamental open problem is instead to determine which deformation classes exist. In each dimension $2n$ only two families are known, $K3^{[n]}$ and Kum_n, along with two sporadic examples, $OG6$ and $OG10$, in dimensions 6 and 10. The guiding conjecture in this direction, the hyper-Kähler SYZ conjecture, was recently proven in a manuscript released by OpenAI [Ope26b], and independently by Engel–Mauri [EM26].

Theorem (Hyper-Kähler SYZ)**.** *Let L be a nontrivial nef line bundle on a hyper-Kähler manifold X which is isotropic for the Beauville–Bogomolov–Fujiki form q_X. Then L is semiample, and a power of L defines a* Lagrangian fibration*: a surjective morphism $f\colon X \to B$ with connected fibers of dimension $n = \frac{1}{2}\dim X$ on which σ vanishes.*

When $b_2(X) \geq 5$, as for all known examples, q_X has isotropic integral classes, so every such deformation class contains a Lagrangian fibration. Combined with the boundedness results of Engel–Filipazzi–Greer–Mauri–Svaldi [EFG$^+$25], it follows that there are only finitely many deformation classes with $b_2 \geq 5$ in each dimension [Ope26b, EM26]. Lagrangian fibrations are therefore the natural gateway to classification. They are also highly constrained: if $f : X \to B$ is a Lagrangian fibration of a hyper-Kähler manifold of dimension $2n$, then its smooth fibers are n-dimensional abelian varieties [Mat99]; the base B is a $\mathbb{Q}$-factorial klt Fano variety of Picard rank one with the intersection cohomology of $\mathbb{P}^n$ [Mat99, SY22], and $B \cong \mathbb{P}^n$ whenever B is smooth [Hwa08, GL14, BS26] or, by a second recent OpenAI manuscript, whenever X is projective [Ope26a], proving a conjecture of Matsushita in the projective case; and f is either isotrivial or of *maximal variation* [Bak25], meaning that the period map $\mathcal{P}\colon B^\circ \to \mathcal{A}_n$, which sends a point of the smooth locus $B^\circ \subset B$ to the isomorphism class of the fiber X_b in the moduli space $\mathcal{A}_n$ of polarized abelian varieties, has n-dimensional image.

The goal of this proposal is to determine where the period map lands, i.e. which abelian varieties, and which families of them, occur as fibers of Lagrangian fibrations. We also hope to use this Hodge-theoretic information to attack structural questions: the classification of Lagrangian fibrations whose fibers decompose (Questions 1 and 2), the extension of the theory to non-compact settings such as Hitchin systems (Questions 3 and 4), and the construction of rational curves and uniruled divisors on hyper-Kähler manifolds (Section 3.3).

2. Past results

My thesis work, supervised by Benjamin Bakker, develops the Hodge theory of Lagrangian fibrations. Let X be a *primitive symplectic variety* of dimension $2n$: a compact symplectic variety with $H^1(X, \mathcal{O}_X) = 0$ and a unique symplectic form up to scaling, such as a hyper-Kähler manifold. Let $f : X \to B$ be a Lagrangian fibration, whose smooth fibers are abelian varieties [Sch20]. Let $f^\circ : X^\circ \to B^\circ$ be the smooth part of f and $V = R^1 f^\circ_* \mathbb{Z}$ the associated variation of Hodge structures, viewed as a representation of $\pi_1(B^\circ)$.

Theorem A ([Var26a]). *If f is not isotrivial, then $V_\mathbb{C}$ is irreducible. If f is isotrivial, then the smooth fibers are isogenous to E^n for an elliptic curve E (first proven in [KLM26]), and $V_K = U_1 \oplus U_2$ splits into two geometrically irreducible summands, where K is the endomorphism field of E.*

Theorem B ([Var26b]). *If f is not isotrivial, then for very general $b \in B$ the Mumford–Tate group of X_b is isogenous over $\mathbb{Q}$ to $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$ for some $h \mid n$. Equivalently, X_b is isogenous to a product $A_1 \times \cdots \times A_{n/h}$ of pairwise non-isogenous h-dimensional abelian varieties, each* Hodge-generic*, i.e. carrying no Hodge classes beyond those forced by its polarization.*

Theorem C ([Var26b]). *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds with $b_2 \geq 7$ which contains a Lagrangian fibration. If $X \in \mathcal{X}$ is general among members admitting a Lagrangian fibration $f\colon X \to B$, then the very general fiber of f is Hodge-generic: $\mathbf{MT}(X_b) = \mathrm{GSp}_{2n}$.*

These results sharpen Bakker's theorem: the period image is not only n-dimensional but Hodge-generic either in $\mathcal{A}_n$ or in a product locus $\mathcal{A}_h^{n/h}$, and in the general case it is the former. The proof of Theorem B hinges on the interplay between the connected monodromy group, which preserves each factor A_i, and the finite part of the monodromy, which by Theorem A must permute the factors transitively. The hypothesis $b_2 \geq 7$ holds for every known deformation type, and by the SYZ theorem every such deformation class contains a Lagrangian fibration.

Personal accomplishments

I've been invited to speak about my research at the Institute for Advanced Study's Research Seminar (November 2025), GTA Philadelphia (May 2026), and the upcoming Special Session on Higgs Bundles at JMM 2027. I've received the Kuga–Sah Award for Excellence in Mathematics from the Stony Brook University Department of Mathematics (2022). I have received the UIC RTG grant for six semesters throughout my time at UIC. I received the Bernard Kurtin Award for research in mathematics from the UIC Department of Mathematics, Statistics, and Computer Science (2025).

3. Research objectives, methods, and significance

We will break our research plan into three primary objectives, which we order by the expected difficulty of the task in question.

RO 1. Determine decompositions of the Mumford–Tate group occur for Lagrangian fibers.
RO 2. Understand the Hodge theory of non-compact Lagrangian fibrations.
RO 3. Work towards the Bogomolov–Mumford conjecture for hyper-Kähler manifolds.

In this section, we will go through each of them, laying out a motivation and an approach for how to address them.

3.1. Which decompositions of the Mumford–Tate group occur?

Theorem B leaves open which values of h actually occur. If $S \to \mathbb{P}^1$ is a non-isotrivial elliptic K3 surface, the induced map $S^{[n]} \to (\mathbb{P}^1)^{(n)} \cong \mathbb{P}^n$ is a Lagrangian fibration whose fibers are isogenous to products of n elliptic curves, so $h = 1$. If singularities are allowed, every h occurs: given a Lagrangian

fibration $Y \to \mathbb{P}^h$ with Hodge-generic fibers (Theorem C), the symmetric power $Y^{(k)} \to (\mathbb{P}^h)^{(k)}$ is a Lagrangian fibration of a primitive symplectic variety with $\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k$. For $h \geq 2$, however, the base $(\mathbb{P}^h)^{(k)}$ does not have the intersection cohomology of projective space (e.g. $\dim \mathrm{IH}^4((\mathbb{P}^2)^{(2)}) = 2$), so it cannot be the base of a Lagrangian fibration of a hyper-Kähler manifold.

Question 1. *Is there a hyper-Kähler manifold (or irreducible symplectic variety) with a Lagrangian fibration such that* $\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$ *with* $1 < h < n$*?*

Question 2. *If X is hyper-Kähler and the very general fiber is not Hodge-generic, i.e. $h < n$, is X of $K3^{[n]}$-type?*

I expect a negative answer to Question 1: whenever the fibers are not Hodge-generic, the fibration should be governed by families of elliptic curves, as it is for $S^{[n]}$. A complete answer would show that decomposable Lagrangian fibrations are exactly those built from elliptic fibrations and give a Hodge-theoretic characterization of the $K3^{[n]}$ deformation type.

Approach. Suppose $h < n$, set $k = n/h$, and pass to the finite étale cover $B' \to B^\circ$ over which V splits as $V_1 \oplus \cdots \oplus V_k$. The symplectic form identifies T_bB with $H^{1,0}(X_b)$, and by Donagi–Markman [DM96] the differential of the period map is a symmetric cubic form on T_bB. Because the period map respects the splitting, this cubic form is a sum of cubic forms on the summands $E_i \subset T_bB$ corresponding to the $H^{1,0}(V_{i,b})$. Consequently each partial period map $\mathcal{P}_i \colon B' \to \mathcal{A}_h$ depends only on the E_i-directions, the E_i integrate to algebraic foliations, and $(\mathcal{P}_1, \ldots, \mathcal{P}_k)$ maps B' dominantly and generically finitely onto a product $Z_1 \times \cdots \times Z_k$ of h-dimensional varieties. This is exactly the structure of $(\mathbb{P}^1)^n \to \mathbb{P}^n$ in the example above. I plan to extend this structure across the discriminant locus and combine it with the constraints on B (Fano, Picard rank one, $\mathrm{IH}^*(B) \cong H^*(\mathbb{P}^n)$, and now $B \cong \mathbb{P}^n$ when X is a projective hyper-Kähler manifold [Ope26a]) and an analysis of the branching of $B' \to B$ to rule out $1 < h < n$. When $h = 1$, I will use the factors to construct an elliptic K3 surface S and compare X with a moduli space of sheaves on S, identifying its deformation type. Recent work of Saccà on compactifications of Lagrangian fibrations [Sac25] will be an important input.

3.2. Lagrangian fibrations beyond the compact case

Many important Lagrangian fibrations are not compact: integrable systems from mathematical physics, local models near singular fibers, and above all the Hitchin fibration $h \colon \mathcal{M}_{\mathrm{Dol}} \to \mathbb{A}$ on the moduli space of Higgs bundles on a curve C [Hit87], which is central to non-abelian Hodge theory and the geometric Langlands program. My compact methods use Bakker's dichotomy, which fails here: h is equivariant for the scaling action $(E, \theta) \mapsto (E, \lambda\theta)$, so its period map is constant along $\mathbb{C}^*$-orbits and never has maximal variation.

Question 3. *Let $f \colon M \to B$ be a proper Lagrangian fibration of a (not necessarily compact) holomorphic symplectic manifold, of maximal variation. Is the Mumford–Tate group of its very general fiber isogenous to* $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h_1} \times \cdots \times \mathrm{Sp}_{2h_k})$ *with* $h_1 + \cdots + h_k = n$*?*

I expect the foliation and monodromy techniques above to extend to this setting, possibly under mild hypotheses on the singular fibers. For the Hitchin fibration of SL_r-Higgs bundles, whose regular fibers are Prym varieties of spectral covers $\Sigma_a \to C$, I can adapt the methods of [Var26a] to show that $R^1h_*\mathbb{C}$ is irreducible over the regular locus $\mathbb{A}^\circ$, despite the failure of maximal variation.

Question 4. *Is the very general* SL_r*-Hitchin fiber Hodge-generic? If not, which products of symplectic groups occur as its Mumford–Tate group?*

Mumford–Tate groups govern the Hodge classes, and conjecturally the algebraic cycles, on the fibers. This complements recent work of Maulik–Shen–Yin on algebraic cycles and Fourier transforms for

Hitchin systems [MSY24, MSY23], and gives a new Hodge-theoretic invariant of a fundamental integrable system.

Approach. Spectral curves form an open subset of a linear system on the compactified cotangent bundle $\mathbb{P}(K_C \oplus \mathcal{O}_C)$, so Lefschetz theory is available: the monodromy is generated by Picard–Lefschetz transformations in vanishing cycles, and Zariski density results [Del74, Lemma 4.4.2[a]] apply. Since the connected monodromy group is a normal subgroup of the derived Mumford–Tate group [And92], showing that the monodromy on the Prym part is Zariski-dense in Sp_{2d}, $d = \dim \mathbb{A}$, would prove Hodge-genericity. The difficulties are that the Hitchin base is only part of a linear system and that the relevant vanishing cohomology is that of the Prym; here we will try to apply the work of Maulik and his collaborators on the topology of Hitchin fibrations [dCMS22, MS24].

3.3. Rational curves and the Bogomolov–Mumford conjecture

By a theorem of Bogomolov and Mumford [MM83], every ample linear system on a K3 surface contains a divisor supported on rational curves. The analogue for hyper-Kähler manifolds is closely tied to the study of algebraic cycles on hyper-Kähler varieties [CMP19]; it is known for the deformation types $K3^{[n]}$ and Kum_n but open in general.

Conjecture (Hyper-Kähler Bogomolov–Mumford)**.** *If X is a projective hyper-Kähler manifold and L is an ample line bundle on X, then $|L|$ contains a uniruled divisor.*

Approach. Let X be general among Lagrangian-fibered members of its deformation class, which exist by the SYZ theorem, so that by Theorem C the period image $\mathcal{P}(B^\circ) \subset \mathcal{A}_n$ is n-dimensional and Hodge-generic. Abelian varieties isogenous to a product $E \times A'$ form the Hecke translates of $\mathcal{A}_1 \times \mathcal{A}_{n-1}$, which has codimension $n-1$, so their expected intersection with the period image is one-dimensional. Results on the existence and density of such typical intersections [KU23, ES25], complementing the Ax–Schanuel theorem [BT19], which controls the atypical ones, should produce a dense set of curves $C^\circ \subset B^\circ$ over which the fibers are isogenous to $E_t \times A'_t$. Over the boundary points of such a curve I expect the elliptic factor to degenerate to a nodal rational curve while A'_t remains an abelian variety, producing uniruled subvarieties in the singular fibers of f. We need to show that these rational curves deform to a general polarized deformation of X, which no longer carries a Lagrangian fibration, using the deformation theory of rational curves on hyper-Kähler manifolds [CMP19]. Then we hope to follow the K3 picture, where an ample class is a combination $a\sigma + bF$ of a section and a fiber. We will try to decompose a divisor in $|L|$ into uniruled horizontal and vertical parts that deform with X.

4. Career development and choice of sponsoring scientist and host institution

My goal is a faculty career at a research university, leading a research program on the Hodge theory of Lagrangian fibrations and its interactions with moduli theory, enumerative geometry, and non-abelian Hodge theory. My training so far is centered on Hodge theory and period maps. I have chosen to pursue the Research Fellowship because full-time research support suits the long-term, higher-risk parts of this program (such as Section 3.3) and the substantial new background they may require. The fellowship will allow me to:

- learn the sheaf-theoretic and enumerative tools that the proposed questions require: perverse filtrations, Fourier transforms on abelian fibrations, moduli of sheaves on K3 surfaces, and Gromov–Witten theory;
- establish my independence, pursuing the more challenging Question 4 and Section 3.3, starting collaborations at MIT and in the Boston area, and traveling to conferences and research visits supported by the fellowship allowance;

- develop as a mentor and expositor through the activities described in Section 5.

My sponsoring scientist, Davesh Maulik, is a leading expert on abelian fibrations, hyper-Kähler varieties, and Hitchin systems, which serve as the backbone of this proposal. His work bears directly on each objective. With Shen and Yin he developed the interaction of Fourier–Mukai transforms with perverse filtrations for abelian fibrations [MSY23], and with Shen, Yin, and Zhang he proved the D-equivalence conjecture for hyper-Kähler varieties of $K3^{[n]}$-type [MSYZ24]. These tools, together with his expertise on moduli of sheaves on K3 surfaces, will help to recognize the hyper-Kähler manifolds in Question 2 as belonging to the deformation space $K3^{[n]}$.

Professor Maulik is a central figure in the topology of Hitchin fibrations, including the proof of the $P = W$ conjecture [dCMS22, MS24] and the study of algebraic cycles on Hitchin systems [MSY24]. His insight will be invaluable to helping attack Question 4. Furthermore, his work on Noether–Lefschetz theory [MP13, KMPS10] and on curves on K3 surfaces [MPT10] sits at the interface of Hodge loci and rational curves that Section 3.3 explores.

MIT offers a large and active community in algebraic geometry and neighboring fields. Moving from Chicago, where I trained in Hodge theory with Bakker, to MIT, with its strengths in moduli theory, enumerative geometry, and geometric representation theory, is precisely the broadening my research program needs.

5. Broader Impacts

Building research communities. As a graduate student I co-organized the Graduate Algebraic Geometry Seminar for two semesters, organized a reading group on the Weil conjectures, and organized a seminar on Lie groups. At MIT I will organize a learning seminar on Lagrangian fibrations and Hitchin systems, open to graduate students and postdocs across the Boston area, and seek support, including from the fellowship allowance, for a small workshop for early-career researchers. Both will be open to everyone and designed to reach researchers who lack local access to this area, including those from groups underrepresented in mathematics. Seminar notes and recorded talks will be posted publicly.

Teaching and outreach. I am currently teaching Calculus II at the University of Chicago as a graduate student lecturer, after five semesters as a teaching assistant at UIC, a Hispanic-Serving Institution where many students are the first in their families to attend college. I have also judged the QED competition of the Math Circles of Chicago; in Boston I will continue this outreach by volunteering with local math circles and competitions such as HMMT, favoring programs that serve schools with limited access to advanced mathematics. I will also seek to mentor undergraduates through the MIT Summer Research Program, which brings students from institutions across the country to MIT to prepare them for graduate study.

Broader scientific impact. Lagrangian fibrations and Hitchin systems are algebraic integrable systems that appear throughout mirror symmetry and mathematical physics; results on their monodromy and periods will be useful well beyond algebraic geometry.

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